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Dear Editor-in-Chief and Editors,

Please find enclosed the manuscript **"Streaming LLM Weights from CXL Memory: A Simulation Study of Regimes, Placement and Pipeline-Aware Pinning"** for consideration in the *Performance Evaluation*. The study has been rebuilt from an earlier version with a new simulator, stronger baselines and new experiments, and is presented explicitly as a simulation and analysis study.

Problem. Models such as Llama-3-70B (141 GB in FP16) exceed GPU memory. CXL Type-3 memory expanders offer cheap host-attached capacity, but a GPU cannot address them: weights reach HBM through host-initiated DMA at roughly 10–20 GB/s on published prototypes. We ask when streaming weights over this path is viable and what a runtime policy can still gain.

Contributions. (1) A host-mediated transfer model, a closed-form arithmetic-intensity threshold, an implementation-independent lower bound and a regime map: streaming is nearly free only above about 16k tokens per iteration and tens of times slower than resident weights for small-batch decode. (2) An analysis of where policy gains come from: placement of the resident layers matters when compute and transfer are comparable (contiguous sets idle the link), and a tuned FlexGen-style per-layer split is already within a few per cent of our policy on dense uniform models. (3) CAMP, a runtime policy (work-conserving prefetcher with next-use admission and stall-cost-aware pinning) that stays within 1.07 of the lower bound in simulation, is up to 2.8× faster than demand paging, ties the best baseline on dense models and is ahead where unit sizes or reuse are non-uniform (up to 1.97× on encoder–decoder inference). (4) A pool-sizing rule: when G GPUs share one expander the threshold rises G-fold, so streaming stays nearly free while $G \leq T/T^{\star}$ (confirmed in simulation up to 16 GPUs). (5) Negative results: the compute-time estimate changes latency by under 1% once calibrated, and LFU is indistinguishable from LRU on cyclic scans. (6) A simulator, cross-checked against a published CXL weight-streaming result, whose link and overlap logic is checked on a real GPU, with the accuracy of that check reported as it is (mixed: 0.6% in link-bound configurations, up to 25% elsewhere).

Fit with the journal. The manuscript models and evaluates the performance of an AI-serving system. It derives a closed-form threshold on the iteration size above which streaming weights from a CXL memory tier is hidden behind computation, an implementation-independent lower bound on iteration latency, and a sizing rule for the number of GPUs one memory expander can feed ($G \leq T/T^{\star}$). It then evaluates scheduling policies with an event-driven simulator whose engine is checked against closed-form pipeline times and whose link and overlap logic is compared with a real GPU. These are the dimensions the journal lists: new modelling and analytic techniques, system sizing and configuration, resource allocation, and simulation methods applied to AI-based services and data-centre memory systems. We position the work against recent literature, including the study of resource allocation for distributed LLM inference by Sun et al. published in *Performance Evaluation* in 2025, with which it shares the combination of validated performance models and a simulator.

Scope of the evidence. This is a simulation study and the manuscript says so in its title, abstract and introduction. No CXL hardware was measured; the CXL link is simulated with parameters anchored in published measurements and the pipeline logic is validated on a PCIe-attached GPU with pinned host memory standing in for the pool. Mixture-of-Experts models, multi-GPU or multi-tenant pools and CXL switches are not evaluated, and the T4 check leaves part of its own error unexplained. Code, raw results and the hardware micro-benchmark are publicly available (see

the Data availability statement).

The manuscript is original, is not under consideration elsewhere, and the authors declare no competing interests. The generative-AI declaration required by the journal is included in the manuscript.

Sincerely,

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